1 **Design & Implementation of Human-Computer Interfaces**
NPTEL-MOOCS

Dr Samit Bhattacharya
Computer Science and Engineering
IIT Guwahati

2 **L4 – Human-Computer Interface Development Life Cycle**

Dr Samit Bhattacharya
Computer Science and Engineering
IIT Guwahati

3 ***Recap***

- What we learned so far
  - General introduction to the field
  - Software engineering life cycles

This lecture – interactive software development life cycle

4 **Waterfall Model (Recap)**

- Seven stages

5 **Waterfall Model and UCD**

- Interactive systems should be designed for the "laymen users" – they should find it "easy to use"

6 **Waterfall Model and UCD**

- The classical waterfall model does not explicitly takes into account this concern
- More focus on efficient "system" design

7 **UCD & Iterative Life Cycle**

- In order to implement UCD, we require highly iterative life cycle
  - Iteration between stages to account for user feedback

8 **UCD & Iterative Life Cycle**

- Waterfall model not suitable

9 **Iteration (too many)**

10 **UCD & Iterative Life Cycle**

- A refined life cycle model is required

11

12 Requirement Gathering

- Should involve end users also, not only “customer/client”
  - Contextual inquiry, ethnographic studies, cultural probes etc

13 Design-Prototype-Evaluate Loop

- Loop required for refinement of early designs based on feedback
- Many iterations
- Should be performed quickly (less time consuming)

14 Design-Prototype-Evaluate Loop

- Design include
  - Interface design (design-prototype-evaluate loop important to take into account user input)
  - Code design

15 Coding & Code Testing

- Can be done in the traditional way
- Early stages and loop important – otherwise costly to rewrite codes and test

16 Empirical Study

- Required to ensure product conforms to usability
- Involves systematic testing with end users (and different from code testing)

17 Key Consideration

- Loops other than design-prototype-evaluate should not be frequent
- Otherwise, development cost (manpower, time, resources) increase

18 Book

- Bhattacharya, S. (July, 2019). Human-Computer Interaction: User-Centric Computing for Design, McGraw-Hill India
  - Print Edition: ISBN-13: 978-93-5316-804-9; ISBN-10: 93-5316-804-X
  - E-book Edition: ISBN-13: 978-93-5316-805-6; ISBN-10: 93-5316-805-8

Chapter 2, Sec 2.1 – 2.4.2